Hassan Khalid

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SUMMARY

AI Engineer with practical experience gained through Jobs and academic projects. Strong foundation in backend development, machine learning, and LLM evaluation, with a passion for building scalable AI applications.

EDUCATION

Forman Christian College University
Bachelor of Science in Computer Science *June 2026*

EXPERIENCE

Single Solution
AI Engineer *Feb 2023 –July 2026*

- Analyzed software engineering issues from **open-source GitHub** repositories and validated fixes using automated test suites.
- Evaluated AI-generated code solutions as part of **LLM benchmarking** workflows, **assessing correctness**, **quality**, and **reliability** across diverse codebases.
- Collaborated with cross-functional teams to identify patterns in AI model failures, providing actionable insights to improve code generation accuracy and reliability.

ACADEMIC PROJECTS

FYP - AI Learning Platform

- Developed an AI-powered learning platform with dynamic quiz generation and personalized coding assessments using **FastAPI**, **Next.js**, and **PostgreSQL**.
- Built RESTful APIs with **async SQLAlchemy**, **JWT authentication**, role-based access control, quiz management, and progress tracking for secure, efficient application performance.
- Created a modular API-driven system connecting the frontend and backend, supporting personalized coding education, scalability, and future AI feature integration.

Social Media Backend API

- Built a social media backend using **FastAPI** with full **CRUD** functionality and **JWT-based** authentication for secure user access control.
- Architected a PostgreSQL database using **SQLAlchemy ORM** and **Alembic**, implementing a post-voting system with composite primary keys.
- Built a **CI/CD** pipeline using **GitHub Actions** and **Docker**, automating integration tests via **Pytest** and production deployment.

Algorithms Implemented from Scratch

- **AI Algorithms:** Linear Regression, Logistic Regression, Decision Trees, K-means Clustering, SVM
- **Sort Algorithms:** Bubble Sort, Insertion Sort, Merge Sort, Quick Sort
- **Search Algorithms:** Breadth First, Depth First, A-Star
- **Other:** Huffman Encoding, Minimax

SKILLS

- **Programming Languages:** Python, SQL, C++
- **Frameworks/Libraries**: NumPy, Pandas, Scikit-learn, Matplotlib, FastAPI, SQLAlchemy, Alembic
- **Tools:** Git/GitHub, GitHub Actions, Jupyter Notebook, VS Code, Cursor
- **Databases:** SQL Server, MySQL, PostgreSQL